Alena Malyarenko

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PROFESSIONAL SUMMARY

- Highly quantitative Environmental Modeller with over eight years of experience developing and executing multi-dimensional fluid dynamics coupled models.
- Proven expertise in developing coupled hydrodynamic and atmospheric models, managing spatial data pipelines via Python, MATLAB, and GIS, and delivering rigorous risk assessments.
- A natural problem-solver who has contributed to multi-million dollar research grants, led international workshops, and peer-reviewed for top-tier journals.
- Expert at translating complex physical data into clear technical publications.
- A Christchurch-based New Zealand permanent resident who thrives independently.

RESEARCH PROJECTS

Current Funding

Program Title:	Antarctic Science Platform Phase Two - Tiaki Moana – What are the drivers and potential implications of unprecedented, climate-driven change in the Ross Sea and Southern Ocean?
Funding Agency:	Hīkina Whakatutuki | Ministry of Business Innovation & Employment, Strategic Science Investment Fund
Role:	Key Researcher
Total Budget:	$49 million
Period:	2025–2032
My Funding:	0.2 FTE per annum, 2027-2032 ($215,000)
My Contribution:	Understanding the oceanic pathways of warm water into ice cavities due to changes in sea ice conditions using the P-SKRIPS model.

Program Title:	Antarctic Sea-Ice Switch - Preparing for New Threats
Funding Agency:	Hīkina Whakatutuki | Ministry of Business, Innovation & Empoyment, Endeavour Fund
Role:	Key Researcher
Total Budget:	$13.6 million
Period:	2024-2029
My Funding:	0.05 FTE per annum ($54,000)
My Contribution:	Investigating how climate change affects the sea ice conditions in the Ross Sea using the P-SKRIPS model.

Program Title:	Grounding zone estuaries: How can ocean stratification, baroclinicity, and tides beneath 500 m of Antarctic ice influence global sea level rise?
Funding Agency:	Royal Society Te Apārangi, Te Pūtea Rangahau a Marsden | Marsden Fund
Role:	Modelling Lead, Postdoctoral Fellow
Period:	2024-2027
My Funding:	0.25 FTE per annum ($120,000)
My Contribution:	Investigating the Ross Ice Shelf cavity conditions with the P-SKRIPS model (MITgcm component); Supervision of PhD students.

Program Title:	Extreme wildfire: Our new reality - are we ready?
Funding Agency:	Hīkina Whakatutuki | Ministry of Business, Innovation & Empoyment, Endeavour Fund
Role:	Key Researcher
Total Budget:	$11.25 million
Period:	2021-2026
My Funding:	0.5 FTE per annum, 2023-2027 ($262,000)
My Contribution:	Developed Weather-FDS: the first atmosphere-fire coupled model that is able to simulate coupled fire-atmosphere interactions

Past Grants

Program Title:	Future of Antarctic Sea Ice Depends on Air-Sea Interactions
Funding Agency:	University of Canterbury Faculty of Science New Idea Seeding Grant
Role:	PI
Total Budget:	$20,000
Period:	2024
My Contribution:	Funded two Research Assistants to contribute to data analysis and Antarctic Energy Balance

Station development; Supplied instrumentation for the Energy Balance Station.

Program Title:	Antarctic Science Platform - The National Modelling Hub
Funding Agency:	Hīkina Whakatutuki | Ministry of Business Innovation & Employment, Strategic Science Investment Fund
Role:	Postdoctoral Researcher
Total Budget:	$49 million
Period:	2018-2025
My Funding:	1 FTE per annum, 2020-2023 ($500,000)
My Contribution:	Developed the P-SKRIPS model for the Ross Sea: the first coupled atmosphere-sea ice-ocean -ice shelf model for that region.

WORK EXPERIENCE

Postdoctoral Researcher
Jul 2023 - present: Te Whare Wānanga o Waitaha | University of Canterbury
Te Kura Aronukurangi | School of Earth and Environment

Adjunct Postdoctoral Researcher
Feb 2020 - present: Te Herenga-Waka | Victoria University of Wellington
Te Puna Pātiotio | Antarctic Research Centre, The National Modelling Hub

Postdoctoral Researcher
Feb 2020 - Jun 2023: Taihoro Nukurangi | The National Institute of Water and Atmospheric Research
Ocean Physics

Researcher
2014 - 2016: Hydrometeorological Centre of Russia, Laboratory of Arctic Hydrometeorology

Researcher
2011 - 2013: Lomonosov Moscow State University, Natural Risk Assessment Laboratory

EDUCATION

PhD (Physics)
2020 - Ōtākou Whakaihu Waka | University of Otago, Te Tari Hū-o-te-Kōhao | Department of Physics
and Taihoro Nukurangi | The National Institute of Water and Atmospheric Research, Te Whanganui-a-Tara — Wellington

MSc in Hydrometeorology (with Distinction)
2015 - Lomonosov Moscow State University, Department of Oceanography, Faculty of Geography

BSc in Hydrometeorology (with Distinction)
2013 - Lomonosov Moscow State University, Department of Oceanography, Faculty of Geography